

Karan Surana

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Profile

Experienced Software Engineer with 3 years of experience building scalable, production grade web applications. Adept at system design and architecture, translating complex requirements into clean and scalable technical solutions.

Education

UNSW Sydney

Masters in Information Technology

Sep 2022 - Sep 2024

Majors: Artificial Intelligence | WAM: Distinction

Sydney

Certification

AWS Certified Cloud Practitioner

AWS Certified AI Practitioner

AWS Certified Solutions Architect Associate

Technical Skills

Languages: TypeScript, Python, JavaScript

Frameworks & Libraries: React, Next.js, Node.js, Express.js, RESTful APIs

Databases & Caching: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Git

Professional Experience

Mid Level Software Engineer | youXpowered

January 2025 - Present

- **Architected and led** a team of 8 in the development of youXinsights, **an enterprise grade**, in-house analytics and visualization platform used by 20+ organisations.
- Built and deployed an in house predictive **Lead Scoring ML Model** using Python, trained on large scale historical application datasets to score new applications accurately.
- Ran requirements discovery with **company stakeholders**, converting ambiguous asks into documented functional and non-functional requirements that drove the platform's design and architecture.
- Engineered **real-time data** visualization infrastructure with advanced cross-filtering and interactive dashboards across **25+ report types**.
- **Solely owned the complete lifecycle** (design, development, deployment, and support) of youXapply, another product, matching customers with lenders, significantly enhancing customer loan conversions.
- **Tech Stack:** TypeScript, JavaScript, Next.js, React, Node.js, AWS.

Full Stack Developer | Gida Technologies

January 2021 - August 2022

- Improved backend development and API integrations for the Antara project, reducing API response time and computational overhead by 60% within the given timeframe.
- Developed a resilient ETL pipeline using Python to efficiently migrate and transform data for downstream services.
- Assisted in frontend development, optimizing API calls and reducing frontend API response time by 40%, improving page load times and user interaction speeds.
- Implemented the complete inventory management system by designing the backend architecture and optimizing database queries through indexing and query optimization techniques.
- Implemented caching mechanisms using Redis to reduce server load and response times.
- Integrated API endpoints with multi-threaded handling for efficient parallel processing of large-scale customer orders, improving system performance by 30%.
- **Tech Stack:** TypeScript, Python, React, AWS, Kubernetes, Redis, Zoho, Shopify, PostgreSQL.

Major Projects

Lucent AI — Natural-Language Data Analytics Platform

[Click here for the link](#)

- Built a production grade natural language querying platform that connects to five database engines — MongoDB, PostgreSQL, MySQL, Microsoft SQL Server and Oracle — letting non-technical users ask questions in plain English and get back answers, charts and dashboards without writing any query.
- Architected the platform so no PII data ever reaches a third-party AI provider.
- Implemented an automated data quality analyzer running seven data quality checks including — type inconsistency, missing fields, null values and more.
- Built an AI-driven visualization layer generating charts across 17 types.
- Created an AI dashboard builder that reads the connected schemas, populates a set of charts, and uses a learning algorithm that generates charts based on the user's preference.

Crypto Spot Trading Bot

- Engineered a production-grade Python/FastAPI trading bot for Bybit and Binance spot markets using CCXT, with two live strategies (trend pullback and mean reversion).
- Designed a risk engine with 7 system states, a 10-gate pre-trade validation pipeline, and dual kill switches to enforce portfolio-level and per-trade safety limits.
- Built a hot-reloadable YAML configuration system with audit logging, SQLite (WAL mode) for trade persistence, and a reconciliation layer for exchange-vs-local state consistency.
- Implemented real-time OHLCV websocket data feeds with regime detection indicators (ADX, Bollinger Bands), including debugging a critical candle-close event bug that locked ADX at 100.